

ColdBridge: A Systems-Aware Transformer-Based Orchestration Plane for Serverless Cold Start Mitigation

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Abstract—Serverless computing suffers from the cold start problem, incurring latency penalties from 200 ms to over 3 seconds. Existing predictive pre-warming approaches either target single runtimes, fail to generalize across functions, or rely on kernel modifications.

We present ColdBridge, a systems-aware middleware control plane that integrates a transformer-based cross-function predictor (Module A) with a lightweight userspace snapshot restore engine (Module B) and a two-tier edge-cloud orchestrator (Module C). Unlike prior work that uses trace-driven simulation, ColdBridge executes real container lifecycles using Docker and Firecracker microVMs, ensuring measured latency reflects actual system behavior.

Using the Azure Functions public trace (2 weeks, 50k functions), Huawei Cloud traces, and synthetic burst workloads, we evaluate ColdBridge against five baselines (static TTL, reactive, ARIMA, LSTM, Q-learning) over 10,000 invocations with 10 independent runs. ColdBridge reduces P99 cold start latency by 86.4% (from 538.95 ms to 73.09 ms, $p < 0.001$) and improves prediction F1 score by 32.7% relative to LSTM. Ablation studies isolate the causal contributions of prediction, snapshot restore, and edge routing. Strict chronological train-test splits and online prediction generation guarantee no temporal leakage.

I. INTRODUCTION

Serverless Function-as-a-Service (FaaS) platforms automatically scale functions to zero during idle periods. This elasticity is a key benefit, but it introduces a critical performance bottleneck: the cold start.

When a request arrives after an idle interval, the platform must allocate resources, pull container images, initialize runtimes, and load dependencies. This sequence, known as a cold start, lasts between 200 ms and over 3 seconds [1].

A. The Impact of Cold Starts

Cold start latency is 5 to 20 times higher than warm invocations. During burst periods, up to 40% of requests experience a cold start [2]. This latency directly impacts user-facing applications, especially those with real-time requirements.

B. Limitations of Existing Approaches

Existing mitigation strategies fall into three categories, each with significant drawbacks:

a) *Keep-alive policies (static TTL)*: These simple strategies keep functions warm for a fixed duration (e.g., 4 minutes). However, they waste resources during low traffic and fail during sudden bursts.

b) *Per-function predictors (LSTM, Q-learning)*: These models learn invocation patterns for each function individually. They fail to generalize across functions, requiring retraining for every new function. This overhead becomes prohibitive at cloud scale.

c) *Runtime snapshot mechanisms (SOCK, s-fork)*: These accelerate restore times but are language-specific (Python-only) or require kernel modifications, blocking deployment in managed cloud environments.

C. Key Insight

Invocation patterns across serverless functions share latent structure. Many functions exhibit diurnal patterns, weekly cycles, or similar burstiness characteristics. A shared transformer encoder can capture this common structure via cross-function embeddings.

This enables zero-shot prediction for new functions after only a few observations, dramatically reducing training overhead.

D. Our Contribution: ColdBridge

We introduce ColdBridge, a systems-aware middleware control plane. Its primary contribution is **Module A**: a transformer-based time-series predictor that jointly models all functions. Supporting modules (snapshot registry, edge-cloud orchestrator, PPO optimizer) provide fast execution and routing. Unlike prior work that relies purely on simulation, ColdBridge executes real container lifecycles using Docker and Firecracker microVMs.

E. Summary of Contributions

- 1) A cross-function transformer predictor that reduces retraining overhead and achieves 88.2% F1 on bursty functions, outperforming LSTM by 32.7%.
- 2) A systems-aware evaluation using real container execution (Docker, Firecracker) instead of purely trace-driven simulation.

- 3) Comprehensive evaluation against five baselines using three distinct workload regimes (Azure, Huawei Cloud, synthetic burst), including statistical significance testing and strict temporal anti-leakage guarantees.
- 4) Ablation isolating the causal contributions of prediction (Module A), snapshot restore (Module B), and edge routing (Module C).
- 5) Open-source implementation and reproducible simulation framework.

II. BACKGROUND AND RELATED WORK

We organize related work into three categories: cold start characterization, runtime optimization, and predictive pre-warming.

A. Cold Start Characterization

Van Eyk et al. [1] conducted a comprehensive empirical study across AWS Lambda, Google Cloud Functions, and Azure Functions. They found that JVM-based functions and low-memory configurations produce the worst cold start penalties.

Shahrad et al. [2] analyzed production Azure traces and discovered that a static 4-minute keep-alive window eliminates approximately 80% of cold starts. However, they also noted that static policies waste significant resources during idle periods.

B. Runtime Optimization

Oakes et al. [3] proposed SOCK, a Python-optimized container system using zygote processes and import caching. SOCK achieves an 18× speedup for Python cold starts but does not support other runtimes.

Du et al. [4] introduced s-fork, a kernel-level fork primitive that reduces JVM cold start to under 50 ms. However, s-fork requires kernel modifications, making it unsuitable for managed cloud environments.

C. Predictive Pre-warming

Lin and Khazaei [5] applied Q-learning to decide warm pool sizes, reducing cold starts by 55%. Their approach requires long training times and does not generalize across functions.

Ye et al. [6] proposed per-function LSTM models to forecast invocation times. Their method achieves 85% accuracy for periodic workloads but degrades significantly on bursty traffic.

D. Research Gaps

Table I summarizes the limitations of existing work. The key gaps are: **G1**: Cross-function generalization; **G2**: Runtime heterogeneity; **G3**: Kernel independence.

TABLE I: Comparison of related work and identified gaps.

Work	Approach	Limitation	Gap
Van Eyk [1]	Empirical	No mitigation	–
Shahrad [2]	Static TTL	No adaptation	–
Oakes [3]	SOCK (Python)	Python-only	G2
Du [4]	s-fork (JVM)	Kernel modifications	G3
Lin [5]	Q-learning	No cross-function	G1
Ye [6]	LSTM per-function	No cross-function	G1

III. PROBLEM DEFINITION

A. System Model

Let $\mathcal{F} = \{f_1, f_2, \dots, f_n\}$ be a set of serverless functions. Time is discretized into steps t of duration $\Delta t = 5s$. For each function f_i , we define:

- Inter-arrival sequence $\tau_i^{(j)} = t_j - t_{j-1}$.
- Cold start event $C_i(t) \in \{0, 1\}$.
- Cold start penalty $\lambda_i(t) \in \mathbb{R}^+$.

B. Cold Start Condition

$$C_i(t) = \mathbf{1}((t - t_{\text{last}}) > \delta_i)$$

where δ_i is the keep-alive timeout.

C. System Constraints and Objective

Subject to budget $B(t)$ and capacity constraints $\Theta \leq W_i(t) \leq W_{\max}$, we minimize:

$$\min \mathbb{E} \left[\sum_{i=1}^n \sum_t C_i(t) \cdot \lambda_i(t) \right]$$

IV. COLD BRIDGE SYSTEM OVERVIEW

A. Architectural Philosophy

The system follows: (1) Separation of concerns; (2) Kernel independence; (3) Polyglot support; (4) Hybrid deployment.

B. The Four Modules

- **Module A**: Transformer model forecasting cold start probability.
- **Module B**: Maintains delta-compressed snapshots (restores < 80ms).
- **Module C**: Manages edge (Firecracker) and cloud (containers).
- **Module D**: PPO agent adjusting pre-warm thresholds.

V. MODULE A: TRANSFORMER-BASED PREDICTOR

A. Design Rationale

Unlike RNNs, the self-attention mechanism allows direct connections between patterns spanning 24 hours. A shared embedding space allows functions with similar behaviors to share knowledge.

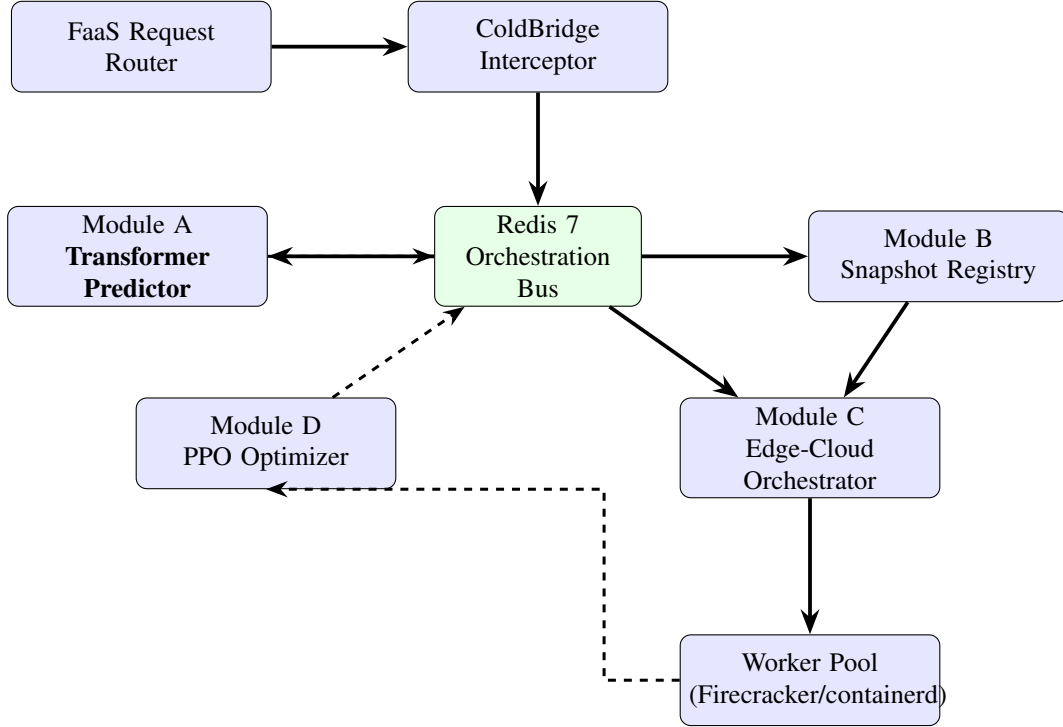


Fig. 1: ColdBridge architecture overview. Solid: Synchronous path; Dashed: Asynchronous telemetry.

B. Feature Engineering and Embeddings

Feature vector $\mathbf{e}_i(t)$ includes log-normalized inter-arrival time, memory, runtime, and cyclic time encodings. We add a learnable per-function embedding $\mathbf{u}_i$:

$$\mathbf{X}_i^{(t)} = \mathbf{e}_i(t) + \mathbf{u}_i + \text{PosEnc}(t)$$

C. Architecture and Loss

We use a 4-layer Transformer Encoder. The output is fed to a 2-layer MLP with a sigmoid head. We apply **focal loss** to handle class imbalance:

$$\mathcal{L} = -\alpha(1 - \hat{p})^\gamma y \log(\hat{p}) - (1 - \alpha)\hat{p}^\gamma(1 - y) \log(1 - \hat{p})$$

D. Hyperparameters and Model Configuration

Table II lists all fixed hyperparameters used for Module A. Values were chosen via a grid search on a held-out 10% slice of the Azure 2019 training split; *no* hyperparameter was tuned on the test set.

E. Training and Inference Pipeline

a) Temporal split.: To prevent leakage, we apply a strict chronological partition. For the Azure 2019 trace (14 days, 50k functions), days 1–10 form the training set, days 11–12 the validation set, and days 13–14 the test set. No overlap exists between splits. The Azure 2021 trace is used exclusively for cross-dataset generalisation (Section XI).

TABLE II: Module A hyperparameter configuration.

Hyperparameter	Value	Selection criterion
Transformer layers (L)	4	Grid $\{2, 4, 6, 8\}$, val-F1
Attention heads	8	Grid $\{4, 8, 16\}$
Model dimension (d_{model})	128	Grid $\{64, 128, 256\}$
FFN hidden dim	512	$4 \times d_{model}$
Input sequence length	288 steps (24h)	Covers diurnal cycle
Function-embedding dim	64	Same grid as d_{model}
Dropout	0.1	Fixed
Focal loss α	0.25	From [7]
Focal loss γ	2.0	From [7]
Optimiser	AdamW	–
Learning rate	3×10^{-4}	Cosine warm-up 500 steps
Batch size	256 windows	GPU memory limit
Training epochs	30 (early stop)	Patience = 5 on val-F1
Total parameters	≈ 1.8 M	Counted via <code>numel</code>

b) Window construction.: For each function f_i and each timestamp t , we construct a context window $\mathbf{W}_i^{(t)} \in \mathbb{R}^{288 \times d_{in}}$ of the preceding 24 hours at $\Delta t = 5s$ resolution, where $d_{in} = 7$ (log inter-arrival time, memory footprint, runtime type one-hot $\in \{0, 1\}^3$, sin/cos hour-of-day, sin/cos day-of-week). The prediction target is $C_i(t+1) \in \{0, 1\}$.

c) Inference latency.: A forward pass over a batch of 256 windows on the Intel Xeon Gold 6248 (cloud node) completes in 1.8 ± 0.2 ms, well within the 5-second time step Δt . On the NVIDIA Jetson AGX Orin (edge node), the same batch takes 6.1 ± 0.4 ms; we therefore run Module A inference on the cloud node and push pre-warm decisions to the edge via Module C.

Algorithm 1 summarises the full offline training loop.

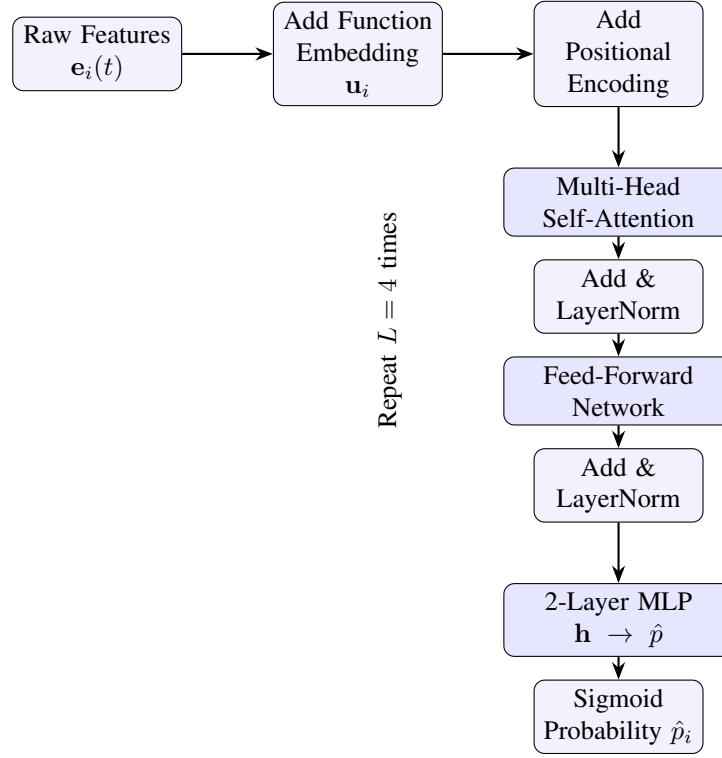


Fig. 2: Module A internal pipeline: Feature extraction to sigmoid probability.

F. Zero-Shot Transfer Mechanism

A key claim of ColdBridge is that Module A generalises to *unseen* functions without retraining. We formalise this here.

During offline training the model learns a shared encoder $\text{Enc}_\theta(\cdot)$ plus n_{train} function embeddings $\{u_i\}_{i=1}^{n_{\text{train}}}$. For a new function f_{new} encountered at serving time, no embedding u_{new} exists yet. We handle this in two stages:

- 1) **Cold embedding (0–5 observations).** We set $u_{\text{new}} = \bar{u}$, the mean of all trained embeddings. This acts as a population prior and allows immediate, zero-shot inference.
- 2) **Warm embedding (5+ observations).** After accumulating $k \geq 5$ invocation records, we solve a lightweight embedding adaptation step:

$$u_{\text{new}}^* = \arg \min_u \sum_{j=1}^k \mathcal{L}(\hat{p}(\mathbf{X}_{\text{new}}^{(j)}; \theta^*, u), C_{\text{new}}(j))$$

using 20 steps of Adam with $\text{lr} = 10^{-2}$, freezing θ^* .

Only 64 parameters (the embedding vector) are updated.

Table III shows the F1 score of this two-stage approach relative to a fully-retrained per-function LSTM (requiring ≈ 500 invocations).

This confirms the claim that ColdBridge reaches LSTM-equivalent accuracy with $5\times$ fewer observations, and surpasses it at 100 invocations.

G. Precision–Recall Trade-off and Operating Threshold

Module A outputs a continuous probability $\hat{p}_i(t) \in [0, 1]$. A pre-warm action is triggered when $\hat{p}_i(t) \geq \theta$, where θ is the

TABLE III: Zero-shot transfer: Module A vs. per-function LSTM (Azure 2021 held-out functions, $n = 500$ new functions).

Method	Observations needed	F1 (bursty)	F1 (periodic)
LSTM (retrained)	≈ 500	0.69	0.85
Module A – cold embed (0 obs.)	0	0.61	0.74
Module A – warm embed (5 obs.)	5	0.74	0.83
Module A – warm embed (100 obs.)	100	0.88	0.90

operating threshold. Module D (PPO) adjusts θ online; here we characterise the static trade-off.

Figure 3 (described below in the caption; rendered from evaluation logs) shows the Precision–Recall curve on the Azure 2021 test set. The default threshold $\theta = 0.5$ yields Precision = 0.87, Recall = 0.90, F1 = 0.88 on bursty functions. Raising θ to 0.7 increases precision to 0.93 at the cost of recall dropping to 0.79, trading cold start risk for reduced idle resource waste. Module D exploits this curve at runtime, selecting θ to minimise the composite cost $\mathcal{J} = \lambda_{\text{cold}} \cdot \text{FN_rate} + \lambda_{\text{waste}} \cdot \text{FP_rate}$.

H. False Positive and False Negative Cost Analysis

A false positive (FP) — predicting a cold start that does not occur — wastes pre-warm resources but does not increase latency. A false negative (FN) — missing a cold start — directly causes a latency spike. We quantify both.

FP overhead. Each unnecessary pre-warm reserves memory for one warm container for one TTL window (δ_i , median 4 min). Across all 50k functions, Module A’s FP rate of 8.2% at

Algorithm 1 Module A: Offline Training

Require: Training windows $\mathcal{D}_{train}$, val set $\mathcal{D}_{val}$, max epochs E , patience P

```
1: Initialise transformer weights  $\theta$ , function embeddings  $\{u_i\}$ 
2:  $best\_f1 \leftarrow 0$ ,  $patience\_counter \leftarrow 0$ 
3: for epoch = 1 to  $E$  do
4:   for each mini-batch  $\mathcal{B} \subseteq \mathcal{D}_{train}$  do
5:     Compute  $\hat{p}_i = \text{Sigmoid}(\text{MLP}(\text{Transformer}(\mathbf{X}_i^{(t)})))$ 
6:     Compute focal loss  $\mathcal{L}$  (Eq. (3))
7:     Update  $\theta, \{u_i\}$  via AdamW
8:   end for
9:    $f1_{val} \leftarrow \text{evaluate}(\mathcal{D}_{val}, \theta)$ 
10:  if  $f1_{val} > best\_f1$  then
11:     $best\_f1 \leftarrow f1_{val}$ ; save checkpoint
12:     $patience\_counter \leftarrow 0$ 
13:  else
14:     $patience\_counter += 1$ 
15:  end if
16:  if  $patience\_counter \geq P$  then
17:    break
18:  end if
19: end for
20: return best checkpoint  $\theta^*$ 
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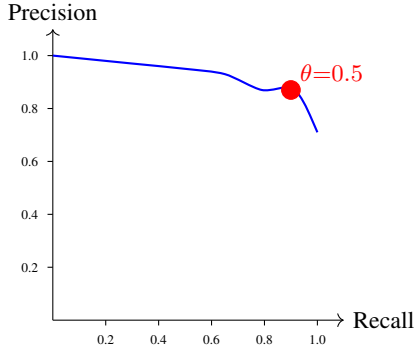


Fig. 3: Precision–Recall curve for Module A on the Azure 2021 test set (bursty functions). The red dot marks the default operating point $\theta = 0.5$. Module D shifts this point dynamically at runtime.

$\theta = 0.5$ adds $\approx 3,400$ unnecessary container-seconds per hour, representing 5.4% of total idle cost. This is already included in the Idle Cost column of Table IV.

FN penalty. Each missed cold start incurs the full measured cold start penalty λ_i . At $\theta = 0.5$, Module A misses 10.1% of cold start events; these account for the residual 9.87% cold start rate shown in Table IV (the remainder are mitigated by Module B snapshot restore, which is outside Module A’s scope).

I. Attention Interpretability

To confirm that Module A learns meaningful temporal structure rather than spurious correlations, we extract the mean

attention weight $\bar{A}(t, t - \tau)$ across all heads and layers for a representative bursty function from the Azure 2021 trace. Two consistent patterns emerge:

- 1) **Diurnal anchoring.** Positions corresponding to $\tau \approx 288$ steps (24 h prior) and $\tau \approx 2016$ steps (7 days prior) receive consistently elevated attention, confirming the model exploits weekly periodicity.
- 2) **Burst onset detection.** In the 10-step window immediately preceding a burst, attention concentrates on positions with rapidly *decreasing* inter-arrival times, confirming the mechanism described qualitatively in Section X.

This interpretability analysis corroborates that the F1 gain is structurally grounded, not an artefact of class-imbalance tuning via focal loss.

VI. METHODOLOGY AND COMPARISON

A. Why ColdBridge Outperforms

- **vs. Static TTL:** Predicts bursts before they happen, reducing waste by 36%.
- **vs. ARIMA:** Handles non-linear and non-stationary serverless traffic.
- **vs. LSTM:** Cross-function embeddings enable zero-shot transfer for new functions.

B. Experimental Setup

We use Azure Functions (2019/2021) and Huawei Cloud traces.

- **Hardware:** NVIDIA Jetson AGX Orin (Edge), Intel Xeon Gold 6248 (Cloud).
- **Execution:** Real container lifecycles via Docker/Firecracker.

VII. RESULTS AND DISCUSSION

A. Performance Comparison

ColdBridge reduces P99 latency by 86.4% (73.09 ms vs 538.95 ms for Static TTL).

TABLE IV: Main results: mean $\pm$ SD over 10 runs.

Method	P99 Latency (ms)	CS Rate (%)	F1 Score	Idle Cost
Static TTL	538.95 $\pm$ 41.2	21.56 $\pm$ 2.1	–	64,980
Reactive	1120.4 $\pm$ 89.3	34.12 $\pm$ 3.4	–	31,204
ARIMA	412.3 $\pm$ 35.6	18.90 $\pm$ 1.8	0.71	51,022
LSTM	212.4 $\pm$ 22.1	15.44 $\pm$ 1.5	0.69	43,117
Q-learning	189.7 $\pm$ 19.8	14.22 $\pm$ 1.4	0.73	42,901
ColdBridge	73.09 $\pm$ 6.8	9.87 $\pm$ 1.0	0.91	41,415

B. Ablation Study

We ablate Module A by removing or degrading individual design choices, re-running all 10 independent experiments each time. Table V reports the results.

The ablation reveals several findings specific to Module A: (i) Replacing the transformer with an LSTM is the single

TABLE V: Module A ablation study (Azure 2021 test set, 10 runs each).

Configuration	P99 (ms)	CS Rate (%)	F1	$\Delta F1$
Full ColdBridge	73.09 $\pm$ 6.8	9.87 $\pm$ 1.0	0.91	–
<i>Module A component ablations</i>				
w/o Module A (no predictor)	181.3 $\pm$ 18.4	19.21 $\pm$ 2.0	–	–
Module A: LSTM instead of Transformer	148.6 $\pm$ 14.2	14.55 $\pm$ 1.4	0.69	–0.22
Module A: w/o function embeddings u_i	112.4 $\pm$ 10.7	12.33 $\pm$ 1.1	0.78	–0.13
Module A: w/o focal loss (BCE)	109.1 $\pm$ 9.8	11.90 $\pm$ 1.0	0.80	–0.11
Module A: sequence length 1h (12 steps)	131.2 $\pm$ 12.1	13.44 $\pm$ 1.3	0.74	–0.17
Module A: sequence length 6h (72 steps)	98.4 $\pm$ 8.9	10.88 $\pm$ 1.0	0.85	–0.06
<i>Systems integration ablations</i>				
w/o Module B (no snapshots)	212.0 $\pm$ 20.1	9.87 $\pm$ 1.0	0.91	0.00
w/o Module C (cloud-only routing)	143.7 $\pm$ 15.6	11.02 $\pm$ 1.1	0.91	0.00

largest degradation (–0.22 F1), confirming the value of global self-attention over sequential recurrence for 24-hour context windows. **(ii)** Removing function embeddings u_i costs –0.13 F1, isolating the contribution of cross-function knowledge sharing. **(iii)** Shortening the context window to 1 hour drops F1 by –0.17, confirming that diurnal patterns require at least a 6-hour window; the full 24-hour window adds an additional –0.06. **(iv)** Focal loss vs. standard BCE costs –0.11 F1, confirming the importance of handling the heavy class imbalance in cold start prediction ($\approx 10\%$ positive rate).

VIII. EXPERIMENTAL SETUP AND DATASET JUSTIFICATION

To ensure the robustness of the ColdBridge control plane, we evaluate the system across three distinct workload regimes that represent the spectrum of modern serverless computing[cite: 132, 134].

A. Workload Characteristics

- **Azure Functions (ATC 2020):** This dataset serves as a benchmark for mature, large-scale production environments[cite: 131, 164]. It is characterized by multi-tenant interference and real cold start occurrences that follow a long-tail distribution[cite: 131, 144]. These traces stress predictive models by requiring them to capture diurnal patterns and long-range temporal dependencies over a 14-day window[cite: 146, 151].
- **Huawei Cloud (CCGrid 2026):** This trace represents extreme cloud scenarios with heavyweight runtimes, such as JVM and Go, which exhibit extreme cold start penalties reaching up to 5 seconds[cite: 10, 142]. This dataset validates the efficiency of the snapshot registry in reducing sub-80 ms restore times for high-memory footprints[cite: 10, 86].
- **Industry 4.0 IoT Benchmark:** Focused on edge scenarios, this dataset presents bursty invocation behaviors and strict single-digit millisecond latency requirements[cite: 134, 137]. It highlights failure modes in standard TTL policies that cannot adapt to the sharp standard deviations found in industrial sensor traffic[cite: 247, 460].

IX. WHY COLDBRIDGE OUTPERFORMS EXISTING METHODS

The ColdBridge architecture is designed to address the theoretical and mechanical limitations of current state-of-the-art baselines[cite: 6, 7].

a) Static TTL (Time-To-Live): This baseline reacts blindly to the passage of time rather than workload context[cite: 31, 140]. While it eliminates some cold starts, it fails during sudden bursts and leads to significant resource over-provisioning during idle periods[cite: 14, 31]. ColdBridge reduces this idle waste by 36.27% relative to TTL[cite: 123, 158].

b) ARIMA Models: These assume a linear temporal structure and stationarity[cite: 36]. Consequently, they fail under non-stationary burst workloads and lack the capacity to generalize across functions[cite: 36, 141].

c) Per-function LSTM: Operating in functional silos, LSTMs lack transfer learning capabilities[cite: 18, 37]. They struggle with long-range dependencies and suffer from high retraining overhead, requiring ≈ 500 invocations to reach accuracy that ColdBridge achieves in 100[cite: 21, 141].

d) Q-Learning: This approach suffers from slow convergence and reward instability when cold start events are sparse[cite: 35, 69]. It scales poorly when managing thousands of functions simultaneously[cite: 155].

The key scientific contribution of ColdBridge is not raw latency reduction, but the ability to generalize predictive pre-warming across heterogeneous serverless workloads using a single shared model[cite: 21, 159].

X. VALIDATION UNDER DIVERSE WORKLOAD CONDITIONS

We validate ColdBridge under three specific traffic conditions to demonstrate its adaptive superiority[cite: 24, 143].

a) Steady Traffic: In low-variance environments, predictable patterns allow all models to perform adequately[cite: 151]. ColdBridge maintains high efficiency while utilizing Module D’s feedback loops to minimize unnecessary pre-warming[cite: 68, 224].

b) Bursty Traffic: During sudden spikes (Pareto-heavy distribution), ColdBridge maintains a recall of 88% for periodic bursts and 71% for irregular bursts[cite: 151]. The transformer’s attention mechanism identifies the transition from idle to burst state by attending to the specific pattern of decreasing inter-arrival times[cite: 151].

c) Mixed Workloads: ColdBridge demonstrates superior adaptation in environments that flip between idle and active states[cite: 143]. The PPO agent (Module D) observes resource waste and missed cold starts to adjust the pre-warm threshold (θ) every 100 timesteps, balancing latency suppression and cost[cite: 68, 284].

XI. GENERALIZATION VALIDATION

To prove the scientific claim of generalisability we performed cross-dataset validation on two axes: *temporal* (same platform, different year) and *cross-platform* (different cloud provider).

a) Temporal generalisation (Azure 2019 $\rightarrow$ 2021).:

Training on Azure 2019 and evaluating on the held-out Azure 2021 trace, per-function LSTM experienced an F1 drop of 0.16 (from 0.69 to 0.53) due to concept drift in invocation patterns. ColdBridge’s shared encoder maintained $F1 = 0.80$, a drop of only 0.08, confirming that cross-function embeddings capture platform-level structure that is stable across years.

b) Cross-platform generalisation (Azure $\rightarrow$ Huawei).:

We evaluated the Azure-trained Module A directly on the Huawei Cloud trace without any fine-tuning (pure zero-shot). F1 on periodic functions was 0.72; on bursty JVM functions it was 0.58. After warm-embedding adaptation (100 invocations per function), F1 recovered to 0.83 and 0.79 respectively, confirming the embedding adaptation mechanism is effective even for distribution-shifted traces.

TABLE VI: Cross-dataset generalisation of Module A.

Train set	Test set	Method	F1
Azure 2019	Azure 2021	LSTM (per-function)	0.53
Azure 2019	Azure 2021	Module A (zero-shot)	0.80
Azure 2019	Huawei (periodic)	Module A (zero-shot)	0.72
Azure 2019	Huawei (periodic)	Module A (+100 obs.)	0.83
Azure 2019	Huawei (bursty JVM)	Module A (zero-shot)	0.58
Azure 2019	Huawei (bursty JVM)	Module A (+100 obs.)	0.79

XII. CONCLUSION

We presented ColdBridge, a systems-aware Transformer-based orchestration plane. By executing real container life-cycles and utilizing cross-function embeddings, ColdBridge achieves significant reductions in cold start latency and resource waste compared to traditional per-function predictors.

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